

5. Functional Requirements

The functional requirements define the operations and features that the system must perform.

User Registration

The system shall allow new users to create an account using their name, email, password, and phone number.

User Login

The system shall allow registered users to log in using their registered email and password.

Product Registration

The system shall allow users to register their products by entering product name, brand name, serial number, warranty start date, and warranty end date.

Product Data Storage

The system shall store user, product, and warranty information in the MySQL database.

Dashboard

The system shall display the total number of registered products, active warranties, expiring warranties, and expired warranties.

Warranty Status Tracking

The system shall automatically determine whether a warranty is active or expired based on the warranty end date.

Warranty Expiry Alert

The system shall display alerts for expired warranties and warranties that are approaching their expiry date.

Warranty Details

The system shall allow users to view registered product and warranty details in an organized table.

Backend API Communication

The system shall enable communication between the React frontend and Spring Boot backend through REST APIs.

6. Non-Functional Requirements

The non-functional requirements describe the quality, performance, and usability characteristics of the system.

Usability

The system should provide a simple and user-friendly interface.

Performance

The system should respond quickly to user actions and database requests.

Reliability

The system should perform consistently without unexpected failures during normal usage.

Security

User login and product information should be handled securely and unauthorized access should be minimized.

Maintainability

The application should have a clear structure so that future modifications and enhancements can be made easily.

Scalability

The system should be capable of handling additional users and products as the application grows.

Availability

The cloud-deployed application should be accessible through a web URL when the services are running.

Compatibility

The web application should work properly on commonly used modern web browsers.

7. Technology Stack

<i>Layer</i>	<i>Technology</i>
Frontend	React.js
Build Tool	Vite
Programming Language	JavaScript
Backend	Java Spring Boot
API	REST API
Database	MySQL
ORM	Spring Data JPA / Hibernate
Database Tool	MySQL Workbench
Development Tool	Visual Studio Code
Version Control	Git & GitHub
Cloud Deployment	Railway

8. Project Modules

The Smart Warranty Portal consists of the following major modules:

Module 1 – User Registration and Login

This module allows users to create an account and securely log in using their registered email and password.

Module 2 – Product Registration

This module allows users to register their products by providing product name, brand, serial number, warranty start date, and warranty end date.

Module 3 – Dashboard

This module provides an overview of the user's registered products and warranty information. It displays total products, active warranties, expiring warranties, and expired warranties.

Module 4-Warranty Management

This module displays complete warranty information for registered products. It automatically determines the warranty status based on the warranty end date.

Module 5 – Warranty Alert and Tracking

This module automatically identifies expired and soon-to-expire warranties and displays appropriate alerts to the user.